



Peltier Module Design For Precision Thermal Management

By understanding the basics of the powerful little Peltier modules, you can better choose the right-sized module with performance ratings to adequately support the thermal management requirements of your design

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Sometimes cooling needs to be controlled more strictly than a simple heat-sink and fan can handle. Rapidly fluctuating loads can make it difficult to keep a component cool and stable. However, this can often be important when trying to improve signal-to-noise performance or prevent damage to a system.

On the other hand, reliability calculations may require a controller or power component to operate below normal ambient temperature. Non-electronic

applications, such as controlling the rate or performance of a chemical reaction, also require precisely managed cooling.

Fortunately, there is a solution to help you tackle tough challenges like these. Thermoelectric modules, or Peltier modules, are available in a variety of sizes that can be connected to an IC package or other types of components in a manner similar to a heat-sink.

Powered by a small current, the module actively draws thermal energy from the attached heat source and dissipates it into the atmosphere. It can be designed to keep the component cooled to a specific temperature, or even below room temperature, if necessary.

A Peltier module comprises two external ceramic plates separated by a series of PN-doped semiconductor pellets that are electrically connected in series. When current passes through the matrix, a temperature gradient is established according to the Peltier effect.

This causes one of the module's boards to cool down, allowing it to absorb heat from a source (such as the surface of a component,

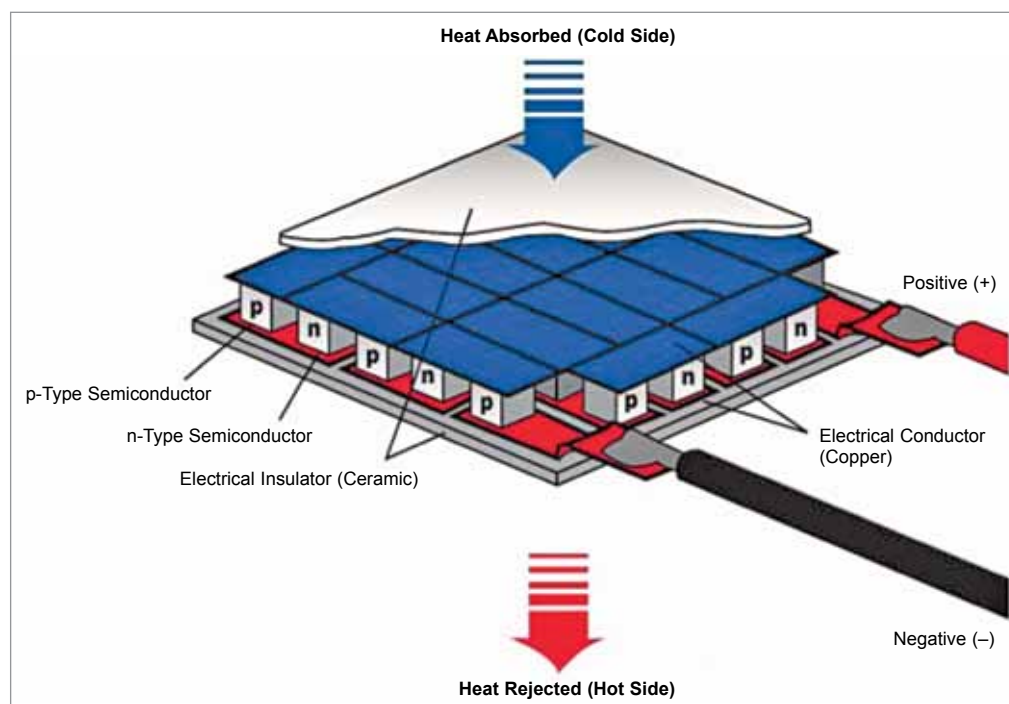


Fig. 1: The constitution of a Peltier module